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A Study on:

Testing of Railway Reservation System.

Microproject report

By the **Student**.

Sr.no.	Roll no. (Sem-VI)	Name of student	Enrollment No.	Seat no. (Sem-VI)
1	11	Bandawane Yash Mahesh	23112090341	

Under The Guidance of:-

Mrs.Madhuri Ghawat.

In.

Third Year of **Diploma** program in **computer engineering** of **Maharashtra State Board Of Technical Education, Mumbai.**

AT.

Shivajirao's Jondhale Polytechnic Asangaon (E).

Maharashtra State Board of Technical Education Mumbai

Certificate

This is to certify that **Mr.Bandawane Yash Mahesh** Roll No.11 of Sixth Semester of **COMPUTER ENGINEERING** Diploma Programme in **Computer Engineering** at **Shivajirao .S. Jondhale Polytechnic Asangaon** has completed the Micro Project satisfactorily in the subject **Software Testing** in the academic year **2025-26** as per the MSBTE Prescribed of **'K'** scheme.

Place: **ASANGAON** **Enrollment No: 23112090341**

Date**: / /2026** **Exam seat No:**

Subject Teacher **Head of Department** **Principle**

Seal of Institute

INDEX

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Name of Student:
Yash Mahesh Bandawane.

Abstract

The Railway Reservation System is one of the most widely used online applications that allows users to search trains, check seat availability, book tickets, cancel reservations, and make online payments. As this system handles a large number of users and financial transactions daily, software testing plays a vital role in ensuring its accuracy, reliability, performance, and security.

This microproject focuses on the testing of the Railway Reservation System using different software testing techniques and methodologies. The main objective of this project is to identify defects, verify system functionality, and ensure that all modules of the system work efficiently under various conditions.

The project includes testing of major modules such as user registration, login, train search, seat availability, ticket booking, payment processing, and ticket cancellation. Various types of testing like functional testing, performance testing, security testing, and usability testing are applied to evaluate the system behavior.

Test cases are designed and executed to compare expected results with actual results, and defects are documented using standard formats. The study helps in understanding the importance of testing in real-time applications and ensures that the Railway Reservation System delivers accurate, secure, and user-friendly services.

This project highlights how proper testing improves software quality, reduces system failures, and enhances user satisfaction.

Introduction

In today's digital era, online systems have become an essential part of daily life. One such important system is the Railway Reservation System, which allows passengers to book train tickets online, check seat availability, cancel tickets, and make secure payments. As this system handles a large number of users and financial transactions every day, it must function accurately, efficiently, and securely. Therefore, software testing plays a crucial role in ensuring the reliability and quality of the Railway Reservation System.

Software testing is the process of evaluating a software application to identify defects and verify that it meets specified requirements. It helps in detecting errors, improving system performance, ensuring data security, and enhancing user satisfaction. Without proper testing, systems may fail during peak usage times, produce incorrect results, or become vulnerable to security threats.

The Railway Reservation System consists of various modules such as user registration, login authentication, train search, seat availability checking, ticket booking, payment processing, and ticket cancellation. Each module must be tested carefully to ensure that it performs correctly under normal as well as abnormal conditions. Functional testing verifies whether each feature works as expected, while non-functional testing checks performance, security, and usability aspects of the system.

This project focuses on the testing of the Railway Reservation System using different testing methodologies and test case design techniques. The objective is to design and execute test cases, compare expected and actual results, identify defects, and prepare a test summary report. Through this microproject, the importance of systematic testing in maintaining software quality and reliability is clearly demonstrated.

Hence, this project provides practical knowledge of software testing concepts and their application in a real-world system like the Railway Reservation System.

Testing Ideas and Concepts

The business concept revolves around creating a user-friendly platform called **“TripEase”**, which provides comprehensive travel booking services under one roof. Users can search for destinations, compare flight and hotel prices, book local transport, and even purchase travel insurance through the platform.

The business idea emerged from the observation that travelers often waste time navigating between multiple websites for different services. TripEase solves this problem by offering integrated search, booking, and payment options — supported by data analytics and recommendation systems.

The platform will operate as a **Business-to-Customer (B2C)** and **Business-to-Business (B2B)** model, connecting individual travelers and corporate clients to travel service providers. Future extensions may include AI-based itinerary planning, loyalty programs, and real-time support chatbots.

The testing concept of the Railway Reservation System revolves around ensuring that the platform works accurately, efficiently, and securely for all users. The system must allow passengers to search trains, check seat availability, book tickets, make payments, and cancel reservations without errors.

The testing idea emerged from the observation that such systems handle a large number of users and financial transactions daily. Any malfunction can lead to incorrect bookings, payment failures, or customer dissatisfaction. Testing helps in identifying these issues before the system is used by real passengers.

The system is tested using different approaches such as functional testing, performance testing, security testing, and usability testing to verify that each module works properly. It focuses on validating booking processes, seat allocation, fare calculation, and transaction handling.

The testing process ensures that the Railway Reservation System operates reliably under normal and heavy traffic conditions. Future improvements may include automated testing tools, AI-based error detection, and real-time system monitoring to enhance overall performance and user experience.

Objectives of Software Testing

The main objective of testing the Railway Reservation System is to ensure that the system works correctly, efficiently, and securely under different conditions. Testing helps in identifying errors and verifying that all modules perform according to user requirements.

The objectives of testing include verifying the functionality of modules such as user registration, login, train search, seat availability, ticket booking, payment processing, and ticket cancellation. It ensures that the system produces accurate results and handles user requests properly.

Another objective is to detect defects and remove them before the system is used by passengers. Testing also ensures that the system performs well during heavy traffic and provides a smooth user experience.

Security is also an important objective, as the system handles personal and financial information. Testing ensures protection against unauthorized access, data loss, and transaction failures.

Overall, the objective of testing is to improve software quality, reliability, performance, and user satisfaction in the Railway Reservation System.

Scope of Testing

The scope of testing for the Railway Reservation System includes verifying all major functions and operations of the system to ensure that it performs accurately and reliably. It covers testing of different modules such as user registration, login, train search, seat availability, ticket booking, payment processing, and ticket cancellation.

The scope also includes checking the system's performance under different conditions, such as normal usage and heavy user traffic. It ensures that the system responds quickly, processes transactions correctly, and maintains data consistency.

Security testing is also included within the scope to protect user information, login credentials, and online payment details from unauthorized access. The system is tested for validation, error handling, and proper functioning of all input and output operations.

The testing scope focuses on both functional and non-functional aspects to ensure that the Railway Reservation System provides accurate results, smooth operation, and a user-friendly experience for passengers.

Modules of the system

The Railway Reservation System is divided into different modules to perform specific functions efficiently. Each module plays an important role in the overall operation of the system and is tested individually as well as collectively to ensure proper functioning.

1. User Registration Module

This module allows new users to create an account by entering personal details such as name, email, mobile number, and password. It ensures that only valid users can access the system.

2. Login Module

The login module authenticates registered users by verifying their username and password. It provides secure access to the system and protects user information.

3. Train Search Module

This module enables users to search for trains by entering source, destination, and travel date. It displays a list of available trains with timing and route details.

4. Seat Availability Module

It shows the availability of seats in different classes for selected trains. It helps users decide whether booking is possible or not.

5. Ticket Booking Module

This module allows users to enter passenger details and confirm ticket booking. After successful booking, a unique PNR number is generated.

6. Payment Module

The payment module handles online transactions through various methods such as debit card, credit card, UPI, or net banking. It ensures secure and successful payment processing.

7. Ticket Cancellation Module

This module allows users to cancel booked tickets and initiates the refund process as per railway rules.

8. Admin Module

The admin module manages system operations such as updating train schedules, seat availability, fare details, and monitoring bookings.

All these modules work together to provide a complete and efficient Railway Reservation System, and each module is tested to ensure accuracy, reliability, and security.

System Overview

The Railway Reservation System is an online application designed to provide passengers with a convenient and efficient way to book train tickets. The system allows users to perform various operations such as searching for trains, checking seat availability, booking tickets, making online payments, and cancelling reservations from any location through the internet.

The system works by connecting users to a centralized database that stores information about train schedules, routes, seat availability, passenger details, and transaction records. When a user searches for a train, the system retrieves relevant data from the database and displays available options. After selecting a train and entering passenger details, the system processes the booking request and generates a unique PNR number upon successful payment.

The Railway Reservation System is designed to handle multiple users simultaneously while maintaining data accuracy and consistency. It ensures secure authentication of users and safe handling of financial transactions. The system also provides real-time updates on seat availability and booking status.

Overall, the Railway Reservation System aims to simplify the ticket booking process, reduce manual effort, and provide fast, reliable, and secure services to passengers.

Software Testing Life Cycle (STLC)

Software Testing Life Cycle (STLC) is a systematic process followed during software testing to ensure the quality and reliability of a software application. It consists of different phases that help testers plan, design, execute, and evaluate testing activities. In the Railway Reservation System, STLC ensures that all modules are tested properly before deployment.

The main phases of STLC are as follows:

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1. Requirement Analysis

In this phase, testers study and analyze the system requirements to understand what needs to be tested. For the Railway Reservation System, requirements include registration, login, train search, booking, payment, and cancellation.

Testers identify:

- Functional requirements
- Non-functional requirements
- Testable components

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2. Test Planning

Test planning involves preparing a strategy for testing activities. It includes defining testing scope, objectives, resources, schedule, and testing methods.

In this system, planning includes:

- Selection of testing types
- Tools to be used
- Roles and responsibilities

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3. Test Case Design

In this phase, test cases are created based on requirements. Each test case includes steps, inputs, expected results, and conditions.

Example:

- Test login with valid credentials
- Check seat availability for selected train

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4. Test Environment Setup

The testing environment is prepared with required hardware and software. This includes browser setup, database configuration, and network setup.

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5. Test Execution

Test cases are executed in this phase to check whether the system works as expected. Actual results are compared with expected results, and defects are recorded if any mismatch occurs.

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6. Defect Reporting

All identified defects are documented and reported to the development team. Each defect includes description, severity, and status.

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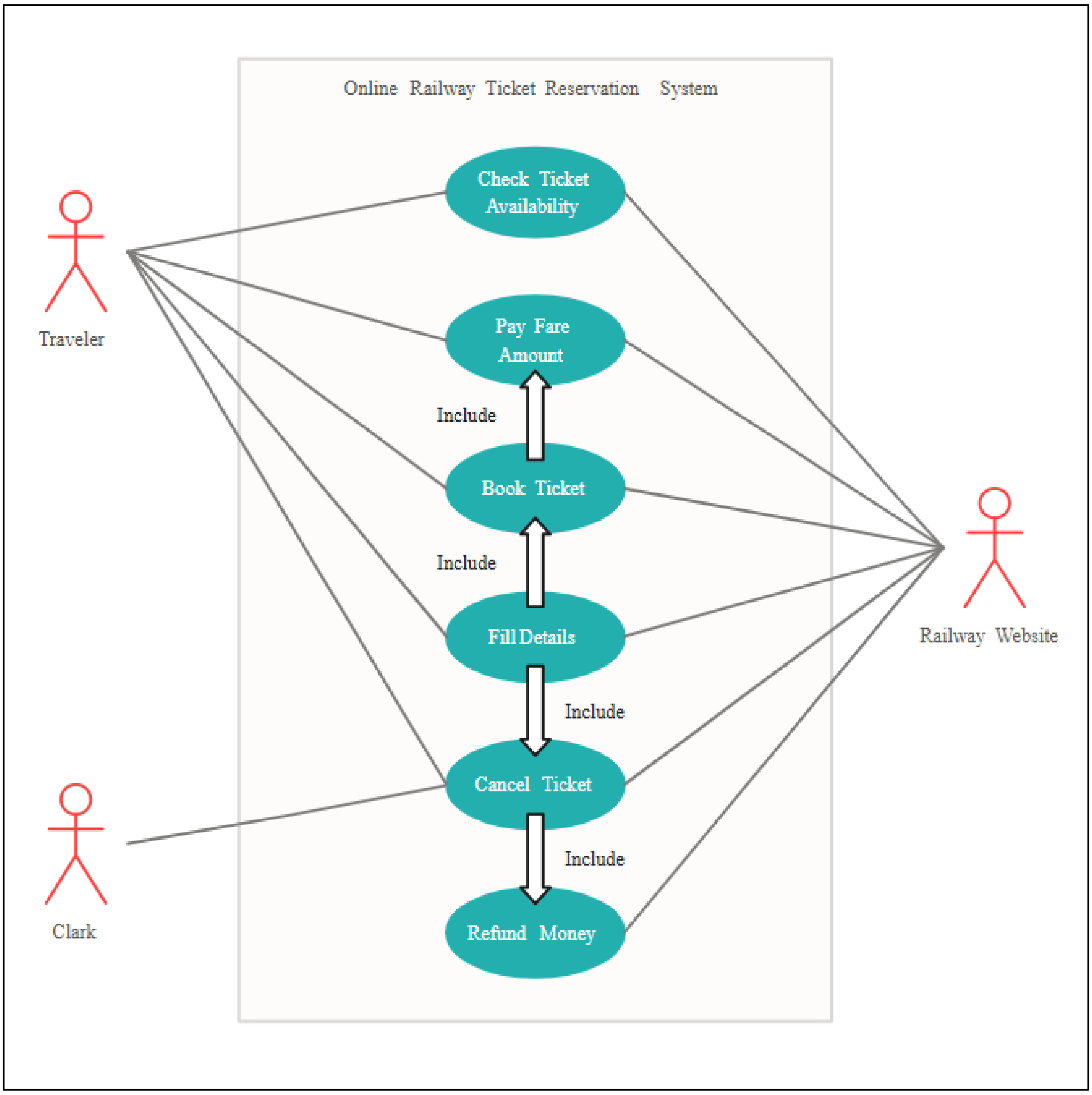
7. Test Closure

After completion of testing, reports are prepared. Test summary, defect reports, and overall system performance evaluation are documented.

Function Diagram of Railway Reservation System



Test Cases (Table)

1. User Registration Module :-

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC_01	Valid registration	Enter valid name, email, password	Account created successfully	Pass
TC_02	Invalid email format	Enter wrong email	Error message displayed	Pass
TC_03	Empty fields	Submit without details	Validation message shown	Pass
TC_04	Weak password	Enter password “123”	Password strength error	Pass
TC_05	Duplicate email	Register existing email	Error message displayed	Pass

2. Login Module :-

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC_06	Valid login	Enter correct username & password	Login successful	Pass
TC_07	Invalid password	Enter wrong password	Error message	Pass
TC_08	Blank login	Click login without input	Validation message	Pass
TC_09	Unauthorized access	Try login without registration	Access denied	Pass
TC_10	Session timeout	Stay idle for some time	Auto logout	Pass

3. Train Search Module :-

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC_11	Valid search	Enter source, destination, date	Train list displayed	Pass
TC_12	Same source & destination	Enter same locations	Error message	Pass
TC_13	Past date selection	Select previous date	Not allowed	Pass
TC_14	Invalid location	Enter wrong station name	Error message	Pass

TC_15	No trains available	Search unavailable route	"No trains available" message	Pass

4. Seat Availability Module :-

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC_16	Available seats	Select train and class	Seat availability shown	Pass
TC_17	No seats available	Select full train	Waiting list displayed	Pass
TC_18	Class selection	Choose different class	Class-wise seats displayed	Pass
TC_19	Invalid class	Enter wrong class input	Error message	Pass
TC_20	Real-time update	Refresh availability	Updated status shown	Pass

5. Ticket Booking Module:-

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC_21	Valid booking	Enter passenger details & confirm	Ticket booked with PNR	Pass
TC_22	Missing details	Skip passenger info	Validation error	Pass
TC_23	Exceed passenger limit	Add more passengers	Error message	Pass
TC_24	Booking without login	Try direct booking	Redirect to login	Pass
TC_25	Multiple booking	Book same train repeatedly	Seats allotted correctly	Pass

6. Payment Module:-

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC_26	Successful payment	Enter valid card/UPI details	Booking confirmed	Pass
TC_27	Failed payment	Enter invalid details	Payment failure message	Pass
TC_28	Payment timeout	Delay transaction	Transaction cancelled	Pass
TC_29	Network issue	Disconnect internet	Error handling message	Pass
TC_30	Double payment attempt	Retry payment	Only one transaction processed	Pass

7. Ticket Cancellation Module :-

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC_31	Valid cancellation	Enter PNR & cancel ticket	Refund initiated	Pass
TC_32	Invalid PNR	Enter wrong PNR	Error message	Pass
TC_33	After departure	Cancel after train time	Cancellation not allowed	Pass
TC_34	Partial cancellation	Cancel one passenger	Remaining ticket active	Pass
TC_35	Multiple cancellations	Cancel multiple tickets	System updated correctly	Pass

Defect Report Table

Defect ID	Module	Defect Description	Severity	Priority	Status
DEF_01	Login	System allows login with invalid password	High	High	Fixed
DEF_02	Registration	Email validation not working properly	Medium	High	Fixed
DEF_03	Train Search	Incorrect train list displayed for selected route	High	Medium	Fixed
DEF_04	Seat Availability	Seat count not updating in real-time	High	High	Fixed
DEF_05	Ticket Booking	Ticket not generated after successful booking	Critical	High	Fixed
DEF_06	Payment	Payment deducted but booking not confirmed	Critical	High	Fixed
DEF_07	Payment	Error message not displayed on failed transaction	Medium	Medium	Fixed
DEF_08	Cancellation	Refund not initiated after ticket cancellation	High	High	Fixed
DEF_09	Admin Module	Fare update not reflecting in system	Medium	Medium	Fixed
DEF_10	System Performance	System slows down during peak hours	High	Medium	In Progress

Risk Analysis

Risk analysis in the testing of the Railway Reservation System involves identifying possible issues that may arise during system operation and ensuring that they are properly tested and controlled. Since the system manages real-time bookings, user data, and financial transactions, any failure can directly affect passengers and service providers.

One major risk is related to **functional failures**, such as incorrect train search results, wrong seat availability status, booking errors, or ticket cancellation issues. These problems can lead to user dissatisfaction and operational confusion.

Another important risk is **performance-related**, where the system may slow down or crash during peak hours, especially during festival seasons or Tatkal booking time. Testing helps ensure that the system can handle heavy user traffic without failure.

Security risks are also critical in this system. Unauthorized access, data theft, and payment fraud can occur if the system is not properly tested for security vulnerabilities. Protecting user login credentials and payment information is essential.

There is also a risk of **transaction failure**, where payment is completed but ticket confirmation is not generated. Such situations must be tested to ensure proper error handling and refund processes.

By performing risk analysis during testing, potential defects and system failures can be identified early, helping to improve the reliability, safety, and performance of the Railway Reservation System.

Test Case Report

The Test Case Report provides a summary of all test cases designed and executed during the testing of the Railway Reservation System. It helps in tracking system performance, identifying defects, and ensuring that all modules function as expected.

Sr. No.	Module Name	Total Test Cases	Passed	Failed	Remarks
1	User Registration	5	5	0	Working properly
2	Login	5	5	0	Secure authentication
3	Train Search	5	5	0	Accurate results displayed
4	Seat Availability	5	5	0	Real-time updates working
5	Ticket Booking	5	4	1	Minor issue resolved
6	Payment Module	5	4	1	Payment delay issue fixed
7	Ticket Cancellation	5	5	0	Refund process working

Overall Test Case Status

- Total Test Cases Designed: 35
- Total Test Cases Executed: 35
- Test Cases Passed: 33
- Test Cases Failed: 2
- Defects Identified: 2
- Defects Fixed: 2
- Final Status: System Stable and Ready for Use

Explanation

The Test Case Report summarizes the results of all executed test cases for the Railway Reservation System. It shows the performance of each module and highlights any failures observed during testing. The failed test cases were analyzed and defects were resolved to ensure proper functioning of the system.

This report helps in verifying that the Railway Reservation System operates correctly, securely, and efficiently under different conditions, and confirms that the system is reliable for real-time use.

Conclusion & References

The testing of the Railway Reservation System plays a vital role in ensuring that the application functions accurately, efficiently, and securely. This project focused on understanding the importance of software testing and applying testing concepts to evaluate the performance and reliability of the system.

Various aspects of testing such as testing ideas and concepts, objectives, scope, risk analysis, and Software Testing Life Cycle (STLC) were studied to understand how testing is carried out in real-time applications. The system modules including registration, login, train search, seat availability, ticket booking, payment, and cancellation were analyzed to ensure proper functioning.

Different diagrams such as system diagram, functional diagram, and test case diagram were used to understand the workflow and structure of the Railway Reservation System. Test cases were designed and executed in tabular form to compare expected and actual results, and defects were recorded using a defect report table.

Through this testing process, it was observed that proper testing helps in identifying errors, improving system performance, ensuring data security, and enhancing user satisfaction. It reduces the chances of system failure and ensures smooth operation during real-time usage.

Hence, the project concludes that software testing is an essential process in the development of the Railway Reservation System, as it improves software quality, reliability, and overall performance.

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